

Research Notes on AlphaGo

AlphaGo

>> Section 1

In the Future of Go Summit held in Wuzhen in May 2017, AlphaGo Master played three games with Ke Jie, the world No.1 ranked player, as well as two games with several top Chinese professionals, one pair Go game and one against a collaborating team of five human players. Google DeepMind offered 1.5 million dollar winner prizes for the three-game match between Ke Jie and Master while the losing side took 300,000 dollars. Master won all three games against Ke Jie, after which AlphaGo was awarded professional 9-dan by the Chinese Weiqi Association. After winning its three-game match against Ke Jie, the top-rated world Go player, AlphaGo retired. DeepMind also disbanded the team that worked on the game to focus on AI research in other areas. After the Summit, DeepMind published 50 full length AlphaGo vs AlphaGo matches, as a gift to the Go community.

>> Section 2

On 29 December 2016, a new account on the Tygem server named "Magister" (shown as 'Magist' at the server's Chinese version) from South Korea began to play games with professional players. It changed its account name to "Master" on 30 December, then moved to the FoxGo server on 1 January 2017. On 4 January, DeepMind confirmed that the "Magister" and the "Master" were both played by an updated version of AlphaGo, called AlphaGo Master. As of 5 January 2017, AlphaGo Master's online record was 60 wins and 0 losses, including three victories over Go's top-ranked player, Ke Jie, who had been quietly briefed in advance that Master was a version of AlphaGo. After losing to Master, Gu Li offered a bounty of 100,000 yuan (US\$14,400) to the first human player who could defeat Master. Master played at the pace of 10 games per day. Many quickly suspected it to be an AI player due to little or no resting between games. Its adversaries included many world champions such as Ke Jie, Park Jeong-hwan, Yuta Iyama, Tuo Jiaxi, Mi Yuting, Shi Yue, Chen Yaoye, Li Qincheng, Gu Li, Chang Hao, Tang Weixing, Fan Tingyu, Zhou Ruiyang, Jiang Weijie, Chou Chun-hsun, Kim Ji-seok, Kang Dong-yun, Park Yeong-hun, and Won Seong-jin; national champions or world championship runners-up such as Lian Xiao, Tan Xiao, Meng Tailing, Dang Yifei, Huang Yunsong, Yang Dingxin, Gu Zihao, Shin Jinseo, Cho Han-seung, and An Sungjoon. All 60 games except one were fast-paced games with three 20 or 30 seconds byo-yomi. Master offered to extend the byo-yomi to one minute when playing with Nie Weiping in consideration of his age. After winning its 59th game Master revealed itself in the chatroom to be controlled by Dr. Aja Huang of the DeepMind team, then changed its nationality to the United Kingdom. After these games were

completed, the co-founder of DeepMind, Demis Hassabis, said in a tweet, "we're looking forward to playing some official, full-length games later [2017] in collaboration with Go organizations and experts". Go experts were impressed by the program's performance and its nonhuman play style; Ke Jie stated that "After humanity spent thousands of years improving our tactics, computers tell us that humans are completely wrong... I would go as far as to say not a single human has touched the edge of the truth of Go."

>> Section 3

On Rotten Tomatoes the documentary has an average rating of 100% from 10 reviews. Michael Rechtshaffen of the Los Angeles Times gave the documentary a positive review and said: "It helps matters when you have a group of engaging human subjects like soft-spoken Sedol, who's as intensively contemplative as the game itself, contrasted by the spirited, personable Fan Hui, the Paris-based European champ who accepts an offer to serve as an advisor for the DeepMind team after suffering a demoralizing AI trouncing". He also mentioned that with the passion of Hauschka's Volker Bertelmann, the film's producer, this documentary shows many unexpected sequences, including strategic and philosophical components. (Rechtshaffen, 2017 John DeFore of The Hollywood Reporter, wrote this documentary is "an involving sports-rivalry doc with an AI twist." "In the end, observers wonder if AlphaGo's odd variety of intuition might not kill Go as an intellectual pursuit but shift its course, forcing the game's scholars to consider it from new angles. So maybe it isn't time to welcome our computer overlords, and won't be for a while - maybe they'll teach us to be better thinkers before turning us into their slaves." Greg Kohs, the director of the film, said "The complexity of the game of Go, combined with the technical depth of an emerging technology like artificial intelligence seemed like it might create an insurmountable barrier for a film like this. The fact that I was so innocently unaware of Go and AlphaGo actually proved to be beneficial. It allowed me to approach the action and interviews with pure curiosity, the kind that helps make any subject matter emotionally accessible." Kohs also said that "Unlike the film's human characters – who turn their curious quest for knowledge into an epic spectacle with great existential implications, who dare to risk their reputation and pride to contest that curiosity – AI might not yet possess the ability to empathize. But it can teach us profound things about our humanness – the way we play board games, the way we think and feel and grow. It's a deep, vast premise, but my hope is, by sharing it, we can discover something within ourselves we never saw before".

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>> Section 4

Algorithm As of 2016, AlphaGo's algorithm uses a combination of machine learning and tree search techniques, combined with extensive training, both from human and computer play. It uses Monte Carlo tree search, guided by a "value network" and a "policy network", both implemented using deep neural network technology. A limited amount of game-specific feature detection pre-processing (for example, to highlight whether a move matches a nakade pattern) is applied to the input before it is sent to the neural networks. The networks are convolutional neural networks with 12 layers, trained by reinforcement learning. The system's neural networks were initially bootstrapped from human gameplay expertise. AlphaGo was initially trained to mimic human play by attempting to match the moves of expert players from recorded historical games, using a database of around 30 million moves. Once it had reached a certain degree of proficiency, it was trained further by being set to play large numbers of games against other instances of itself, using reinforcement learning to improve its play. To avoid "disrespectfully" wasting its opponent's time, the program is specifically programmed to resign if its assessment of win probability falls beneath a certain threshold; for the match against Lee, the resignation threshold was set to 20%.